

VEDANT SAHAI

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EDUCATION

Pennsylvania State University

May 2024

Master of Science in Computer Science & Engineering (GPA: 3.75 / 4.00)

- Teaching Assistant: CMPSC 132: Programming and Computation II: Data Structures

University of Mumbai

May 2021

Bachelor of Engineering in Computer Engineering (CGPA: 9.58 / 10.00)

EXPERIENCE

Distyl AI Inc.

April 2025 – Present

AI Engineer

- Increased Day-by-Day Prior Auth AI decision throughput by over 50% for multi-date cases, by parallelizing stay and bed level Guideline Picker AI system using LangGraph-based orchestration.
- Engineered sub-10-minute SLA Prior Auth AI processing, by designing a FastAPI orchestration layer with Procrastinate-backed background execution for deterministic LLM workflows.
- Executed a zero-data-loss migration of 2 critical environments (UAT & PLAT) with minimal downtime, delivering an F50 client AI platform landing-zone transition on containerized infrastructure.
- Enabled image-to-AI-system automation on Distyl's platform by designing a prompt-orchestrated GenAI pipeline that converts raw customer images into deployable, platform-native AI systems.

Julep AI Inc.

Oct 2024 – April 2025

Machine Learning Engineer

- Increased User Engagement by 15% within a month of launch by developing the "Browser Use" feature, enabling automation of browser tasks through AI-driven workflows.
- Optimized the workflow performance to accelerate executions from hours to under 45 minutes for 2.5k runs by migrating the database from CozoDB to TimescaleDB.
- Developed Julep Python-based CLI and 10+ AI workflow templates, improving developer participation to 5k GitHub stars.
- Enhanced platform features by 20% by programming a FastAPI-Docker service enabling interaction between AI workflows and External Tools.

JP Morgan Chase & Co.

Jun 2023 – Aug 2023

AIML Summer Associate

- Improved the Transaction Detection Rate (TDR) by 100-150 basis points by implementing the Online ML XGBoost algorithm for the Transaction Risk model (TRS).
- Attained 90% accuracy by programming a TabNet-based Deep Neural Network as a challenger for the TRS XGBoost model.
- Ensured 95% code coverage by implementing a PyTest-based testing framework for the TRS Feature Engineering codebase.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, Java, LaTeX

ML Frameworks: PyTorch, TensorFlow, RASA, Scikit-learn, XGBoost, HuggingFace, NLTK, SpaCy, LlamaIndex, Langchain, Langgraph

Databases: MySQL, PostgreSQL, MongoDB, Timescale

Cloud: Amazon Web Services (AWS), EC2, S3, Lambda, API Gateway, SageMaker, IAM

Technologies: React.js, Flask, FastAPI, PyTest, GitHub Actions, Helm, Kubernetes, Git, Docker, PySpark, FERN, Sphinx, Temporal, Procrastinate

PROJECTS

Datacertus

May 2022

Reacts. JS, AWS APIs, Python, Docker, AWS ELB, AWS Lambda, DynamoDB, Scikit, RASA, PyTorch, Keras, Scikit-Learn

- Trained a BERT summarization model to summarize updates on disaster information with 75% ROGUE-L.
- Increased update frequency by over 50% by developing a pipeline using Lambda, Selenium, and the BERT model to track natural calamities.
- Reduced data processing time by over 10% by integrating AWS Lambda, S3, and API Gateway-based trigger events.

Conversational AI for Secure Healthcare Assistance

May 2021

Docker, Node.JS, MongoDB, RASA, Blockchain, Python, Express.JS, jQuery, AJAX, Vault, Nginx

- Processed over 10,000 Electronic Health Records concurrently by architecting a microservice-based software.
- Attained the intent prediction confidence up to 95% by developing and integrating a RASA-powered therapy chatbot.
- Integrated the BigchainDB + IPFS blockchain database with the RASA for behavior analysis and secure medical record storage.

PUBLICATION

Vedant S., et al. (2021) Leveraging Deep Learning and IoT for Monitoring COVID19 Safety Guidelines Within College Campus. In: Garg D., Wong K., Sarangapani J., Gupta S.K. (eds) Advanced Computing. IACC 2020. Communications in Computer and Information Science, vol 1367. Springer, Singapore.